

# SituationReport

**get\_data**

## User Manual

Export Jira data for SituationReport

---

situation-report -- [github.com/Jaegerfeld/situation-report](https://github.com/Jaegerfeld/situation-report)

For Agile Coaches and PI Managers -- Version 0.9.5

# 1 Exporting Data from Jira Cloud

**Note:** The `get_data` module is still under development. This manual describes the manual export of Jira data via the Jira REST API. This is the same process that `get_data` will automate — the exported JSON files can be processed directly by `transform_data`.

This chapter explains how to export real issue data from Jira Cloud for SituationReport. The export produces the same JSON structure as the Testdata Generator — `transform_data` processes both identically.

## 1.1 What you need

| Requirement        | Details                                             |
|--------------------|-----------------------------------------------------|
| Jira Cloud account | An account with read access to the desired project  |
| API token          | Personal token from id.atlassian.com — created once |
| Project key        | The Jira project shortcode, e.g. 'ART_A' or 'SCRUM' |
| curl or browser    | curl for automated queries; browser for quick tests |

## 1.2 Creating an API token

The API token replaces your password for API calls and is created once:

| Step | Action                                                                                                        |
|------|---------------------------------------------------------------------------------------------------------------|
| 1    | Open <a href="https://id.atlassian.com">https://id.atlassian.com</a> in your browser (while logged into Jira) |
| 2    | Security → API tokens → Create API token                                                                      |
| 3    | Enter a label, e.g. 'SituationReport', click 'Create'                                                         |
| 4    | Copy the token and store it safely — it is only shown once!                                                   |

*Security note: Treat the API token like a password. Store it in a password manager and do not share it.*

## 1.3 Querying the Jira REST API

Jira Cloud offers two API variants. Use the current v3 API for new scripts; existing v2 scripts continue to work. `expand=changeLog` is **required in both** variants — without it, status transitions are missing and `transform_data` cannot compute time-in-stage values.

### Current API (v3) – recommended

Endpoint: `POST /rest/api/3/search/jql`

Parameters are passed as a JSON request body. The response additionally includes `isLast` and `nextPageToken` for pagination (Section 1.5).

```
curl -X POST \  
  "https://company.atlassian.net/rest/api/3/search/jql" \  
  {  
    "jql": "project = ART_A",  
    "startAt": 0,  
    "maxResults": 50,  
    "expand": "changeLog"  
  }
```

```
-u "name@company.com:YourAPIToken" \
-H "Content-Type: application/json" \
-d '{"jql":"project=ART_A ORDER BY created ASC",
    "maxResults":100,
    "expand":["changelog"],
    "fields":["issuetype","created",
              "status","summary"]}' \
-o ART_A_page1.json
```

*maxResults is limited to 100 issues per request with the v3 API. Use nextPageToken for projects with more than 100 issues (Section 1.5).*

### Older API (v2) – still supported

Endpoint: GET /rest/api/2/search

Parameters as URL query string. Supports up to 1,000 issues per request and offset-based pagination with startAt.

```
curl -u "name@company.com:YourAPIToken" \
     "https://company.atlassian.net/rest/api/2/search?jql=project=ART_A+ORDER+BY+created+ASC\
&expand=changelog\
&maxResults=1000\
&fields=issuetype,created,status,project,summary,resolution" \
-o ART_A_page1.json
```

*Tip: For new scripts, the v3 API is preferred. Existing v2 scripts do not need to be migrated.*

## 1.4 Fields required by transform\_data

| Jira field                   | Required | Purpose                                                     |
|------------------------------|----------|-------------------------------------------------------------|
| key                          | ✓        | Issue identifier (e.g. ART_A-001)                           |
| fields.issuetype.name        | ✓        | Issue type for filtering (Feature, Bug, ...)                |
| fields.created               | ✓        | Issue creation date                                         |
| fields.status.name           | ✓        | Current status                                              |
| changelog (expand=changelog) | ✓        | Status transitions with timestamps — MUST be set in the URL |
| fields.resolution            | –        | Optional — resolution type (e.g. 'Done', 'Won't Fix')       |
| fields.summary               | –        | Optional — issue title                                      |

## 1.5 Pagination – more than 100 issues

### API v3 – Cursor-based pagination (nextPageToken)

The v3 API returns at most 100 issues per request. Each response contains either a nextPageToken for the next page, or "isLast": true when all issues have been fetched. Pages must be retrieved **sequentially** — the token from the current response is used in the next request.

```
# Page 1 – no nextPageToken
```

```

curl -X POST \
  "https://company.atlassian.net/rest/api/3/search/jql" \
  -u "name@company.com:Token" \
  -H "Content-Type: application/json" \
  -d '{"jql":"project=ART_A ORDER BY created ASC",
      "maxResults":100,"expand":["changelog"],
      "fields":["issuetype","created",
               "status","summary"]}' \
  -o ART_A_page1.json

# Read nextPageToken from page 1:
#   cat ART_A_page1.json | grep nextPageToken
#   (or with jq: jq -r '.nextPageToken' ART_A_page1.json)

# Page 2 - insert nextPageToken
curl -X POST \
  "https://company.atlassian.net/rest/api/3/search/jql" \
  -u "name@company.com:Token" \
  -H "Content-Type: application/json" \
  -d '{"jql":"project=ART_A ORDER BY created ASC",
      "maxResults":100,"expand":["changelog"],
      "fields":["issuetype","created",
               "status","summary"],
      "nextPageToken":"<token from page 1>"}' \
  -o ART_A_page2.json

# Repeat until isLast:true appears in the response

# Merge all pages
python -m helper ART_A_page1.json ART_A_page2.json ... \
  --output ART_A_merged.json

```

Repeat until the response contains "isLast": true. The **Helper** merges all pages and removes duplicates automatically.

### ***API v2 – Offset-based pagination (startAt) – Legacy***

The v2 API supports up to 1,000 issues per request. With `startAt`, any page can be accessed directly.

**Important:** `startAt` is only compatible with the v2 endpoint — not with the new v3 endpoint.

```

# Page 1 (issues 1-1,000)
curl -u "name@company.com:Token" \
  "https://company.atlassian.net/rest/api/2/search?\
jql=project=ART_A&expand=changelog&maxResults=1000&startAt=0" \
  -o ART_A_page1.json

# Page 2 (issues 1,001-2,000)
curl -u "name@company.com:Token" \
  "https://company.atlassian.net/rest/api/2/search?\
jql=project=ART_A&expand=changelog&maxResults=1000&startAt=1000" \
  -o ART_A_page2.json

# Merge files with the Helper

```

```
python -m helper ART_A_page1.json ART_A_page2.json \  
--output ART_A_merged.json
```

Increment startAt by 1,000 until the response contains fewer than 1,000 issues.

## 1.6 From JSON export to workflow file

After exporting, create a workflow file that maps your actual Jira statuses. The following Python snippet extracts all status names from the export:

```
import json  
data = json.loads(open('ART_A_page1.json').read())  
statuses = set()  
for issue in data['issues']:  
    for h in issue['changelog']['histories']:  
        for item in h['items']:  
            if item['field'] == 'status':  
                statuses.add(item['toString'])  
print(sorted(statuses))
```

From the status names found, create a workflow.txt — arrange the statuses in logical order and set the <First> and <Closed> markers. The workflow file format is described in the **transform\_data** User Manual.

## 2 Frequently Asked Questions (FAQ)

---

### What happens if `expand=changelog` is missing?

`transform_data` finds no status transitions and outputs empty time values for all issues. The Excel files contain no usable data. Always include `expand=changelog` in the URL.

### I get an 'Unauthorized' error when calling curl.

Check: Is the email address correct? Is the API token correct? Are they separated by a colon? Does your account have read access to the project? Tip: Generate a new API token if the old one was lost or expired.

### Can I merge multiple Jira projects?

Yes — either via JQL (`jql=project in (ART_A, ART_B)`) for a single export, or merge two separate exports with the Helper. Make sure the workflows of both projects are compatible.

### How large can the JSON file get?

Rule of thumb: approximately 2–5 KB per issue (depending on changelog length). 1,000 issues  $\approx$  2–5 MB. For very large projects (10,000+ issues), use pagination and the Helper.

### Should I use test data or real data?

For demos, training, and installation tests, synthetic test data from the **Testdata Generator** is the better choice — no privacy concerns, reproducible, and fully controlled. For real flow analyses and decisions, actual Jira data reflects the true workflow state.

## 3 Glossary

---

| Term         | Explanation                                                                                |
|--------------|--------------------------------------------------------------------------------------------|
| ART          | Agile Release Train — a team-of-teams structure in SAFe                                    |
| API          | Application Programming Interface — programming interface                                  |
| API token    | Security key for accessing the Jira REST API; replaces the password in curl calls          |
| Changelog    | Jira log of all status changes for an issue (requires expand=changelog)                    |
| CFD          | Cumulative Flow Diagram — shows the cumulative issue count per stage over time             |
| Closed stage | The stage that marks an issue as completed (workflow.txt: <First>Closed<First>)            |
| Cycle time   | Time from the first active stage (<First>First<First>) to the closed stage                 |
| First stage  | First active stage where cycle time measurement begins (workflow.txt: <First>First<First>) |
| Helper       | SituationReport module for merging multiple Jira JSON files                                |
| Issue        | A work item in Jira (Feature, Bug, Enabler, Story, ...)                                    |
| JQL          | Jira Query Language — query language for Jira searches, e.g. project=ART_A                 |
| JSON         | JavaScript Object Notation — format of Jira API exports                                    |
| Stage        | One step in the workflow (corresponds to a Jira column or status name)                     |
| Workflow     | Ordered sequence of stages that an issue passes through                                    |